AI-Powered Research Gap Identifier & Trend Analyzer

R26-SE-023

Introduction

The Problem

Students and early-stage researchers struggle to identify a novel research topic, find the right supervisor/collaborators, and track research progress in a structured way. Current processes are manual, slow and often lead to topic duplication, poor supervision fit, and delays.

Users & Stakeholders

- Students / Researchers
- Academic supervisors
- Research coordinators
- Universities and research institutions

Introduction

- **Impact**
  - **For Students:** Difficulties with finding unique research ideas, identifying research gaps, and difficulties searching supervisors & guidance.
  - **For Supervisors:** Difficulty tracking multiple projects, Lack of centralized monitoring tools, Having difficulties managing and communicating with students.
  - **For Institutions:** Reduced research originality, Weak transparency in research governance

- **Credible Evidence**
  - Sri Lanka: only 25.39% of eligible students entered the national university system in 2021 .
    https://nhrdc.gov.lk/nhrdc/media/attachments/2024/07/23/statistical-bulletin---2023_edition4.1_compressed.pdf
  - PhD completion varies and is not uniformly high; supervision fit and support are widely recognized as critical factors.
    https://cgsnet.org/wp-content/uploads/2022/01/phd_completion_and_attrition_analysis_of_baseline_demographic_data-2.pdf
  - Many institutions manage supervisory load formally, showing workload feasibility is a real constraint.
    https://ukcge.ac.uk/assets/resources/Managing-Supervisory-Loads-Centrally-Limiting-Student-Numbers-Taylor-UKCGE.pdf

Introduction

Existing Solutions and Their Limitations

- Manual selection: slow, biased toward well-known supervisors
- Keyword match tools: miss deeper relevance and ignore workload
- Basic recommenders: weak use of collaboration networks; poor handling of new topics and supervisors.

The Proposed Solution

AI-Powered Research Gap Identifier & Trend Analyzer with 4 modules:

1. Intelligence Engine: S.M.D.M. Bandara
2. Collaboration Engine: K.L. Widanagama
3. Operations Workspace: L.P.P. Wijesinghe
4. Integrity & Governance: Darshan R.

How this overcomes the limitations :

Automates novelty checking, gives feasible supervisor matches, supports end-to-end research execution, and improves transparency through logging/provenance.

Introduction

https://youtu.be/wPRNfH3B6V4

Presentation Introduction Video

Generated using HeyGen AI Video Generator

IT22141842 – MUDITHA BANDARA



Semantic Analysis & Gap Detection

Contribution:
Takes the retrieved similar papers, extracts their limitations and uses LLM reasoning to validate if the user's raw proposal fills a genuine research gap.

Novelty:

- Evaluates user-submitted research ideas predictively at the ideation stage.
- Performs semantic gap detection and indentify novelties using LLM reasoning.

SEMANTIC ANALYSIS & GAP DETECTION

- **Novelty:**
  - **Problem Definition**
    - **Information Overload:** Rapid growth - research gaps increasingly difficult.
    - **System Limitations:** Retrospective tools - Semantic Scholar, Elicit (They find what exists, not new, unformulated ideas/ literiture).
    - **Manual Inefficiency:** Finding papers is time-consuming, error-prone, and challenging.
  - **Main Research Objective**
    - Develop a Similarity-Guided Semantic Gap Detection Engine that uses NLP and LLM reasoning to determine whether a proposed research idea addresses a genuine research gap.
  - **Sub-objectsives**
    - Extraction evidence from relevant literature.
    - Generate semantic representations of research ideas and literature.
    - Perform semantic alignment and identify gaps.
    - Apply LLM-based reasoning for gap & novelty assessment

SEMANTIC ANALYSIS & GAP DETECTION

- **Application of Domain Knowledge**
  - **Natural Language Processing (NLP):** For extracting and preprocessing text from academic papers.
  - **Semantic Similarity Analytics:** Vector embeddings enable conceptual comparison between research ideas and literature.
  - **Transformer Architecture:** To capture contextual relationships in scientific text.
  - **LLM Reasoning:** Synthesize evidence across multiple research papers.

- **Research Methodology**
  - **Similarity-Guided Retrieval:** Identify research papers most related to the proposed idea.
  - **Section-Level Extraction:** Automated parsing of PDFs to isolate Methodology, Limitations, and Future Work.
  - **Semantic Representation:** Mapping the user idea and extracted text into a shared Vector Space.
  - **Semantic Alignment:** Compare research ideas with identified research limitations.
  - **LLM Gap Reasoning:** Classify the idea (as Unique Gap, Partial Overlap, or Low Novelty).

SEMANTIC ANALYSIS & GAP DETECTION

- **Latest Technologies Utilized**
  - **Embedding Models:** SciBERT/ Sentence-BERT/ DeBERTa-v3 - for embeddings
  - **LLM :** GPT-4o, LLaMA-3, Claude 3.5 - for high-level reasoning.
  - **Vector Databases:** Pinecone/ Milvus for high-speed semantic similarity searches.
  - **Parsing:** Grobid/ PyMuPDF - for structural extraction of academic PDF sections.
  - **Data Sources:** arXiv, Semantic Scholar Open Research Corpus

- **Validation & Evaluation Metrics**
  - **Similarity Accuracy:** Cosine similarity and embedding evaluation for gap classification.
  - **Extraction Performance:** Precision, Recall, F1-Score
  - **Expert Validation:** Comparing system-generated gap reports against expert reviews.
  - **User Study:** Usefulness and efficiency for early-stage researchers.

- **Data Security & Ethical Clearance**
  - **Data Integrity:** Ensure Fair Use when parsing published PDFs.
  - **Security:** Encrypted API endpoints for research idea submissions.
  - **Ethics:** Mitigate AI Hallucinations by using evidence-focused prompts.

SEMANTIC ANALYSIS & GAP DETECTION

- **System Overview**
  - Analyzes user-submitted research ideas
  - Extracting evidence from related literature
  - Perform semantic comparisons with prior studies
  - Applying LLM reasoning to evaluate research novelty
  - **Output:** Generate an interpretable gap assessment report.

- **Dependencies and Common Components**
  - Academic literature Sources (arXiv, Semantic Scholar)
  - Transformer embedding models (SciBERT, SBERT, DeBERTa)
  - Large Language Models (GPT-4 / LLaMA / Claude)
  - NLP processing libraries and document parsing tools

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Recommendation & Collaboration Network -

Provides the system's supervisor recommendations for a research idea, helping users quickly find the right academic support network

Novelty - Uses a graph neural network on the collaboration network and includes workload feasibility to avoid recommending overloaded supervisors.

Recommendation & Collaboration Network

Novelty

- Uses a collaboration graph and GNN to recommend supervisors.
- Filters out overloaded supervisors using workload feasibility.

Problem Definition

- Students struggle to find the right supervisor for a research idea.
- Topic-only methods can recommend supervisors who are not feasible.

Main Objective

- Recommend Top-K supervisors that are both relevant and feasible.

Sub-objectives

- Build datasets and representations.
- Build collaboration graph and apply GNN.
- Rank, filter by workload, and compare with baselines.

Recommendation & Collaboration Network

Domain Knowledge

- Recommender systems
- Graph networks (collaboration & supervision links)
- Workload/capacity-based allocation

Research Methodology

- Design Science: build the component
- Experimental: compare with baselines

Latest Technology Used

- Inductive heaterogenous graphSAGE Graph Neural Network

Recommendation & Collaboration Network

Data Availability

- Data: synthetic, supervisor profiles, expertise, group links, collaboration/co-supervision, supervision history, workload indicators
- Security: role-based access, protect submitted research text
- Ethics: data minimization, anonymize IDs for experiments, institutional approval if real records are used

Evaluation

- Compare baselines vs GNN and workload to show better relevance and feasibility
- Metrics: Precision@K, Hit@K (recommendation quality)
- Overload-rate@K (feasibility/workload filtering)

Recommendation & Collaboration Network

System Overview

- Store profiles and research ideas in a database
- Auto-build/update the collaboration graph
- Generate node features from text and metadata
- GNN scores supervisor–idea matches
- Apply workload filtering
- Output Top-K supervisors

Dependencies

- Shared user/profile management (accounts, roles)
- Shared database with common schema (profiles, ideas, relationships)
- Shared API/auth layer (secure access, logging)

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Collaborative Workspace & Progress Tracking

Contribution to the Overall Solution

- Provides a centralized digital workspace for research collaboration
- Enables structured task management and milestone tracking
- Supports real-time progress monitoring for supervisors and students
- Improves communication and coordination in research projects

Novelty

- Integrates collaboration, task management, and progress tracking into a single platform
- Designed specifically for academic research environments
- Provides visual dashboards for research progress monitoring
- Eliminates the need for multiple disconnected tools

Collaborative Workspace & Progress Tracking

Novelty

- Centralized collaborative workspace for research projects
- Integrated progress tracking and milestone monitoring
- Real-time dashboards for project visualization
- Platform tailored for academic research supervision

Problem Definition

- Research projects involve multiple activities such as task coordination, document sharing, and communication.
- These activities are often managed using disconnected tools.
- Supervisors struggle to monitor multiple student projects effectively.
- Lack of transparency leads to communication gaps and project delays.

Research Objective

Main Objective

Design and develop a collaborative workspace and progress tracking system that enables students and supervisors to manage and monitor research activities through a centralized digital platform.

Sub-Objectives

- Create a collaborative workspace environment
- Implement research task and milestone tracking
- Develop dashboards to visualize research progress
- Support structured communication between students and supervisors

Collaborative Workspace & Progress Tracking

Research Methodology

- Design Science Research (DSR) approach
- Focuses on creating and evaluating a technological artifact
- Research stages:
  - Identifying the research problem
  - Analyzing system requirements
  - Designing the system architecture
  - Implementing the prototype
  - Evaluating the proposed solution

Several Knowledge Domains Used

- Software Engineering
- Collaborative Systems
- Web Application Architecture

Technologies Used

- React.js for frontend interface
- Go (Golang) with Gin framework for backend services
- Supabase cloud platform with PostgreSQL database

Collaborative Workspace & Progress Tracking

Validation and Evaluation

System Evaluation Methods:

- Technical testing
- User-based evaluation.

Performance Metrics:

- System response time
- Progress tracking accuracy
- System reliability
- User satisfaction

Data Availability, Security and Ethics

Data Sources:

- Literature review
- System testing feedback from students and supervisors

Security Measures:

- Role-based access control
- Secure authentication
- HTTPS communication
- Controlled access to project data

Ethical Considerations:

- Protection of user privacy
- Voluntary participation in system evaluation
- No sensitive personal data collection

Collaborative Workspace & Progress Tracking

System Overview

The web-based Collaborative Workspace Component provides:

- Research project workspace creation
- Task and milestone management
- Progress tracking dashboards
- Document sharing and management
- Supervisor feedback and communication
- Notification and alert system

Dependencies and Common Components

Dependencies

- Authentication service
- Workspace management module
- Task tracking module
- Progress monitoring module
- Notification service

Common Components Shared with System

- User management
- Database services
- API gateway
- Cloud storage

IT22097156 – DARSHAN R



Integrity & Governance

Contribution

- Acts as the "trust layer" for the main Gap Analyzer system.
- Ensures that only unique, verified, and tamper-proof data enters the broader research mapping workflow.

Novelty

- Real-time Pre-Commit Semantic Filtering (SBERT + FAISS).
- Event-driven append-only research ledger with role-aware governance tracking.

INTEGRITY & GOVERNANCE

- Novelty
  - **Problem Definition**
    - **Verifiable provenance tracking** of research activities
    - **Transparent contributor accountability**
    - **Detection of semantically overlapping research topics**
    - **Integrated governance mechanisms** for approvals and milestones

  - **Research objective**
    - Develop an **Integrity & Governance Module** that ensures:
      - **Tamper-evident tracking** of research lifecycle events
      - **Transparent contributor accountability**
      - **Semantic novelty detection** to prevent redundant research topics

    - Sub-Objectives
      - Design a **hash-linked append-only ledger** for research events
      - Implement **contribution mapping** linking actions to user identities
      - Integrate **SBERT + FAISS** for semantic similarity detection
      - Provide **DAG-based visualization** of research provenance

INTEGRITY & GOVERNANCE

- **Application of Domain Knowledge**
  - Blockchain & Distributed Ledger Technology
    - → Tamper-evident provenance tracking
  - Natural Language Processing (NLP)
    - → Semantic similarity detection using SBERT
  - Vector Search & Information Retrieval
    - → FAISS indexing for fast similarity queries
  - Data Provenance Models
    - → W3C PROV and DAG-based research lineage representation

- **Research Methodology – Knowledge domain that contributes to methodology**
  - What specific Latest Technology/Technologies in that Domain are utilized to develop the proposed solution
  - How to Validate and Evaluate the Proposed Solution
  - Data Availability, Security and Ethical Clearance

- **Technologies Used**
  - Sentence-BERT (SBERT) — semantic embedding of research topics
  - FAISS — high-speed vector similarity search
  - Smart Contracts (Solidity) — governance rule enforcement
  - DAG Visualization (NetworkX / D3.js) — research lineage tracking

INTEGRITY & GOVERNANCE

System Validation

The proposed solution will be evaluated using:

- Semantic Detection Accuracy
- System Performance
  - Semantic similarity response time (<2 seconds)
  - Ledger transaction throughput
- Integrity Verification
  - Hash-link validation to detect tampering

Data Sources

- Open academic repositories (arXiv, CrossRef, Semantic Scholar)
- Simulated research topic submissions

Security & Ethical Considerations

- User submissions anonymized during evaluation
- No permanent storage of sensitive text during similarity checks
- Compliance with institutional research ethics guidelines

INTEGRITY & GOVERNANCE

Implementation Details

- **System Overview of the Individual Component**
  - **Semantic Validation Layer -** Detects semantically similar research topics before submission
  - **Integrity Ledger -** Hash-linked append-only record of research lifecycle events
  - **Provenance DAG -** Visual representation of research history and contributor actions
  - **Smart Contract Governance -** Enforces approval workflows for supervisors and committees

- **Dependencies and common components**
  - SBERT + PyTorch for semantic embeddings
  - FAISS for similarity indexing
  - Blockchain-style ledger storage for provenance

- **Outcome**
  The module provides:
  - Tamper-evident research provenance
  - Automated governance enforcement
  - Semantic novelty detection for research topics

This strengthens transparency, accountability, and innovation in academic research systems.

AI-Powered Research Gap Identifier & Trend Analyzer

Commercialization & Investment

Estimated Development Investment

Cost Category	Estimated Cost (LKR)
Human Development Effort (4 developers)	240,000
Cloud Infrastructure & Deployment	80,000
Testing, Dataset Preparation & Misc	20,000
Total Initial Investment	**≈ 350,000 LKR**

Cost Efficiency

- System built using **open-source technologies**
- Utilizes **free-tier cloud infrastructure** for experimentation
- Personal resources reduce hardware investment

- **Revenue Model**
- Software-as-a-Service (SaaS) – Monthly subscription for institutions
- Institutional Licensing – Annual license for universities
- API Integration – Paid access for integration with academic platforms

AI-Powered Research Gap Identifier & Trend Analyzer

Market, Value & Cost Recovery

Target Market
- Universities and higher education institutions
- Research institutes and innovation labs
- Corporate R&D departments

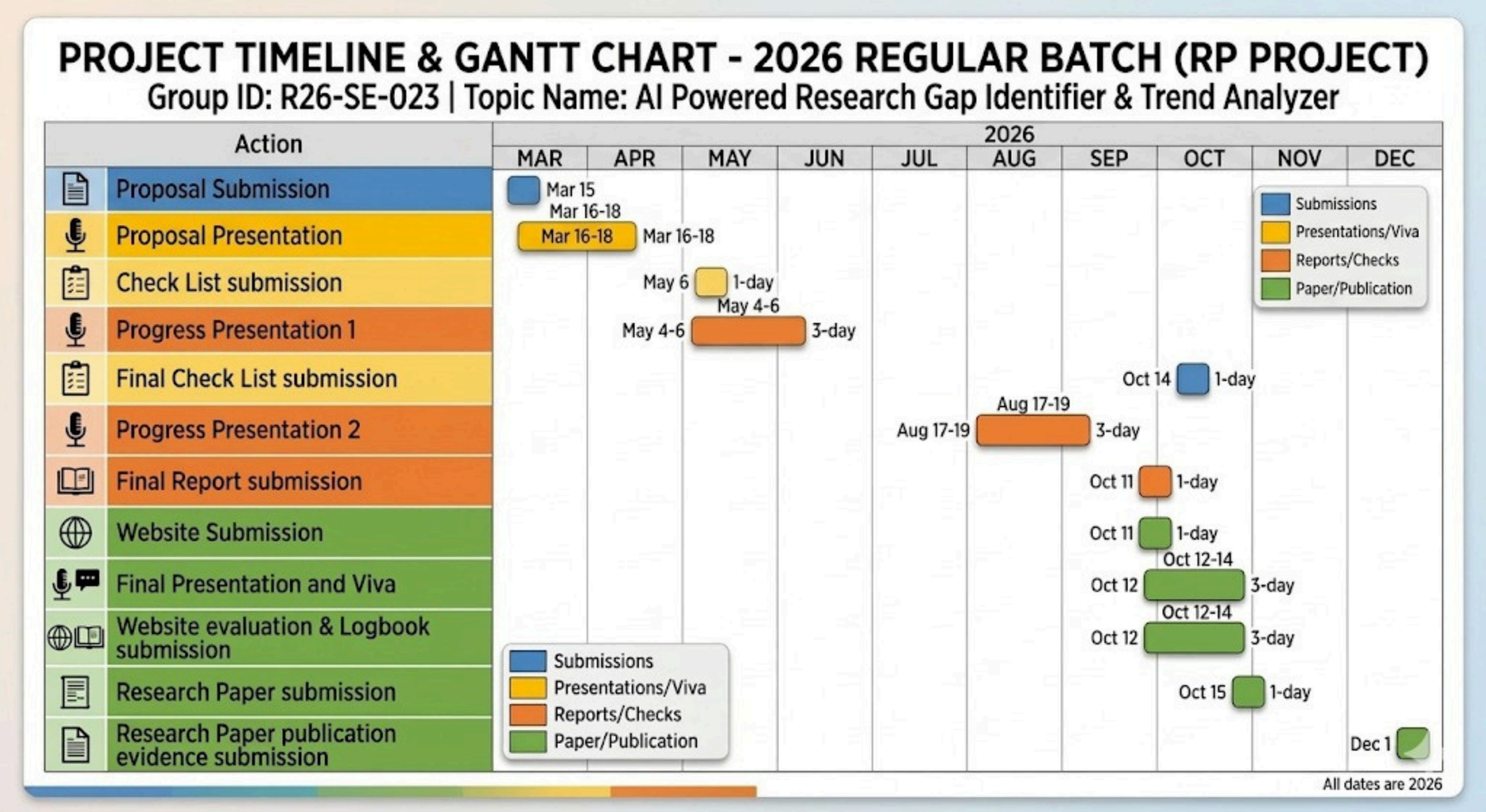
Customer Profile
- Research supervisors and academic administrators
- PhD / MSc researchers and students
- University research management offices

Value Proposition
- AI-powered research gap detection and semantic analysis
- Supervisor recommendation and collaboration insights
- Research integrity and provenance tracking
- Reduces time spent on manual literature analysis

Intellectual Property
- Proprietary algorithms for semantic analysis and recommendation
- Copyright protection for system architecture and software
- Institutional licensing agreements

Cost Recovery
- Academic Subscription Example: 50,000 LKR / month
- Initial investment (~350,000 LKR) can be recovered within ~7 months with one institutional client.

AI-Powered Research Gap Identifier & Trend Analyzer



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